ERIK SANGEORZAN

C++ Systems Engineering | Embedded Android & Audio

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Portfolio: ejsang.github.io/erik-sangeorzan-shure-portfolio/
Michigan | Open to Niles, IL hybrid work and relocation
Authorized to work in the U.S.; no sponsorship required

SUMMARY

Software engineer with 4+ years of professional C++ systems experience at Ford, including a six-month embedded Android/audio rotation, plus prior automotive acoustics experience at HARMAN. Professional ownership spans networking, native debugging, automated validation, and hardware/software integration.

PROFESSIONAL EXPERIENCE

Ford Motor Company | Software Engineer

Mar. 2022 - Present

Dearborn, MI

Current assignment: Core Diagnostic Protocol Requirements & Tools

- Own and extend a multithreaded C++ DoIP edge-node simulator and vehicle communications library for automotive diagnostics, protocol validation, and engineering integration.
- Implemented TCP recovery, diagnostic timing APIs, 29-bit CAN support, and TLS handshake, configuration, and hostname verification, with regression tests.
- Diagnose protocol and native-code failures using Wireshark, logs, and focused test builds; fixed a diagnostic memory leak and socket lifecycle/recovery issues.
- Added remote simulator control and reset capabilities that helped reduce a major automated test-suite runtime by 61% in 2025; helped raise library line coverage from 10.8% to 53.2% in 2025.
- Configured CMake/Ninja for local and CI builds; migrated copied dependencies to NuGet and delivered repeatable x86/x64 packages, quality checks, and release documentation.
- Integrated SonarQube across three repositories and onboarded four engineers; mentor colleagues in TDD and code quality, participate in peer code reviews, and facilitate sprint planning.

IVI Audio | Ford College Graduate rotation

Sep. 2022 - Mar. 2023

- Built and flashed customized AOSP images on Qualcomm IVI development hardware using Linux, ADB, and Fastboot in an Android/QNX environment.
- Instrumented existing native C audio-test code and adjusted vendor configuration; investigated Audio HAL/PAL startup and playback using custom debug output and logs.
- Modified AudioReach/QACT graphs and A2B/TDM routing; achieved audio playback and recording on development and custom IVI hardware, working with supplier engineers and documenting validation procedures.

FCG rotation sequence: Core NetCom Diagnostics; IVI Audio; ADAS Localization/Odometry; ADAS Embedded Data Operations. Returned to Core NetCom after the program.

HARMAN International, Automotive

May - Aug. 2021

Acoustic Engineering Intern | Novi, MI

- Built a C# amplifier electrical-validation tool adopted for production testing; reduced a five-to-eight-hour manual workflow to less than one hour of human effort.
- Performed electrical/acoustic documentation, vehicle-audio benchmarking, and critical listening to investigate frequency-response, distortion, level, and phasing concerns.

SELECTED ACADEMIC EMBEDDED & AUDIO PROJECTS

Embedded Audio Separation of Human Voices

Six-person academic team | C/C++, Python, Teensy, Raspberry Pi Linux, I2S/UART

Team built an embedded audio capture, streaming, ICA-processing, and playback system. Implemented windowing and splicing algorithms for continuous processing; demonstrated separation of linear mixtures.

Stratum Embedded Synthesizer

Four-person academic team | E100 processor, Altera DE2-115, assembly

Team built a musical synthesizer with sampled instruments and layer recording/playback. Contributed layering, additional peripheral-driver routines, and memory implementation.

STM32L4 Zumo Autonomous Navigation

Two-person academic team | C, STM32 HAL, ADC/GPIO, PWM, LPUART

Team implemented autonomous obstacle-course navigation toward a light source, combining light/distance sensing with PWM motor control. A separate polling line-following extension used six reflectance sensors, weighted motor correction, and UART telemetry.

TECHNICAL SKILLS

Professional systems: Modern C++, C#, TCP/IP, UDP, DoIP, CAN, TLS, socket programming, Wireshark

IVI Audio rotation: AOSP image builds, ADB/Fastboot, Linux CLI, Android/QNX, native C debugging, AudioReach/QACT, A2B/TDM

Engineering practices: Git, CMake/Ninja, Visual Studio, SonarQube, NuGet, unit/integration testing, CI; TDD, SOLID/OO design, Scrum

Academic projects: C/C++, Python, assembly, STM32L4, Teensy, Raspberry Pi, GPIO, UART/I2S, PWM, DSP/ICA

EDUCATION

University of Michigan, College of Engineering Dec. 2021

B.S.E. Computer Engineering, Minor in Music, magna cum laude

CERTIFICATIONS

Ford Clean Code Green, Orange, Yellow and White Belts - internal certifications

LEADERSHIP & SERVICE

Ann Arbor Jaycees | Community Vice President (2026); Director (2025)

Member since Dec. 2023; help plan and run Ann Arbor's July 4th Parade and coordinate group service supporting food access, mental health, foster care, parks, and animal welfare.

Michigan Drumline | Section Leader, 2020-2021